

MANIKANTA SURYASAI SUNKARA

AI/ML Engineer | Full-Stack Developer | Blockchain Developer
Andhra Pradesh, India | +91 7893562841 | smanikanta713@gmail.com

LinkedIn: linkedin.com/in/manikanta-suryasai-sunkara-60955b27b | GitHub: github.com/Manirider

PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer and Full-Stack Developer with hands-on experience building production-grade machine learning pipelines, NLP systems, and blockchain applications. Proficient in Python, Java, Solidity, and TypeScript with demonstrated expertise in MLOps, deep learning, and decentralized systems. Contributed 171+ commits across 31 public repositories spanning AI/ML, Backend Engineering, MLOps, and Web3. Passionate about building scalable, secure, and high-performance software solutions in Agile environments.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, Solidity, SQL

AI/ML & NLP: TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM, BERT, DistilBERT, Sentence Transformers, SHAP, Optuna, ONNX Runtime, Hugging Face Transformers

MLOps & Data: MLflow, Docker, Kubernetes, CI/CD (GitHub Actions), Prometheus, Grafana, Celery, SMOTE, MICE Imputation

Web & Backend: FastAPI, Flask, Spring Boot, Next.js, React, REST APIs, Microservices, Uvicorn

Blockchain & Web3: Solidity, Hardhat, Ethers.js, OpenZeppelin, ERC-20, ERC-721, UUPS Proxy, AMM/DEX, DAO Governance

Cloud & Databases: AWS (Basics), PostgreSQL, MySQL, Redis, Firebase, SQLite

Tools & Practices: Git, Docker Compose, Agile/Scrum, Unit Testing (Pytest, Mocha), Code Review, JIRA, Power BI, Problem Solving, Team Collaboration, Communication

PROFESSIONAL EXPERIENCE

Full-Stack Developer Intern

Anurag IT Solutions | Jan 2025 - Jun 2025 | Hyderabad, India

- Developed and deployed Spring Boot REST APIs handling 5,000+ requests/day, reducing average response time by 30% through query optimization and caching strategies.
- Containerized 3 microservice applications using Docker and Docker Compose, improving deployment reliability and reducing environment setup time by 70%.
- Collaborated with cross-functional teams in an Agile/Scrum environment, participating in daily stand-ups, sprint planning, and code reviews to ensure code quality.
- Wrote comprehensive unit and integration tests achieving 85%+ code coverage using JUnit, improving overall system reliability.

AI/ML Intern

APSCH (Andhra Pradesh State Council of Higher Education) | Mar 2024 - Apr 2024

- Built and evaluated machine learning models (Logistic Regression, Random Forest, XGBoost) for real-time prediction tasks, achieving 92% F1-score on validation data.
- Performed exploratory data analysis, feature engineering, and hyperparameter tuning using Scikit-learn and Optuna to optimize model performance.
- Documented model performance metrics and presented findings to stakeholders, enabling data-driven decision-making.

Data Analytics Intern

THUB (Telangana Hub for Entrepreneurship) | May 2024 - Jun 2024

- Created interactive Power BI dashboards visualizing KPIs across 10+ datasets, enabling real-time insights for leadership.

- Automated data extraction and transformation pipelines using Python and SQL, reducing manual reporting effort by 60%.
- Cleaned and preprocessed large datasets (50K+ records) to ensure data integrity and accuracy for downstream analytics.

PROJECTS

Multitask NLP Platform | Python, PyTorch, FastAPI, ONNX, MLflow, Docker, Prometheus

- Engineered a production-ready multi-task NLP API serving Sentiment Analysis, NER, and Question Answering via a shared DistilBERT encoder, reducing memory footprint by 60% compared to separate models.
- Optimized inference latency to under 50ms on CPU by exporting to Int8 quantized ONNX models (260MB to 65MB), with full MLflow experiment tracking and Prometheus observability.

Healthcare Outcome Prediction | Python, Scikit-learn, XGBoost, LightGBM, SHAP, Streamlit, Docker

- Built an end-to-end ML pipeline predicting patient treatment outcomes with 0.92 F1-score and 0.99 AUC-ROC, featuring MICE imputation, SMOTE, Optuna hyperparameter tuning, and SHAP interpretability.
- Quantified cost-effectiveness savings of approximately \$700K/year per 1,000 patients through model-guided treatment allocation with interactive Streamlit dashboard.

Omnichain Asset Bridge | Solidity, Hardhat, Node.js, Ethers.js, SQLite, Docker

- Developed a production-quality, two-chain asset bridge implementing Lock/Mint and Burn/Unlock patterns with cross-chain governance, crash recovery, replay protection, and RBAC using OpenZeppelin.
- Built a Node.js relay with SQLite persistence ensuring exactly-once delivery semantics, 28 unit tests, and Docker-orchestrated multi-chain infrastructure.

End-to-End MLOps Platform | Python, FastAPI, MLflow, Docker, PostgreSQL, Redis, Celery, Kubernetes

- Automated model training, experiment tracking, versioning, and deployment using MLflow, Celery distributed task queues, and Kubernetes manifests, reducing manual deployment effort by 50%.
- Implemented 52 tests (unit + integration), Prometheus/Grafana monitoring, API key authentication, and CI/CD via GitHub Actions for a fully production-grade platform.

Multi-Agent AI System | Python, Groq, Google Gemini, OpenAI

- Designed a multi-agent orchestration system with Researcher, Writer, and Critic agents collaborating via shared state, deterministic workflows, and structured observability logging.

LLM Security Middleware | Python, Flask

- Built a defense-in-depth security middleware protecting LLM applications from prompt injection, jailbreaks, and data leakage, achieving 90%+ attack detection rate with less than 5% false positives.

EDUCATION

B.Tech in Artificial Intelligence and Machine Learning

Aditya College, Surampalem | 2024 - Expected Jun 2027 | CGPA: 8.80/10

Diploma in Computer Science and Engineering

Godavari Institute of Engineering and Technology | 2021 - 2024 | Percentage: 84%

ACHIEVEMENTS

- Maintained 31 production-grade public GitHub repositories across AI/ML, Backend, MLOps, Blockchain, and Security domains with 171+ contributions in the past year.
- Reduced organizational workflow time by 60% using SharePoint automation, improving team productivity.
- Core Entrepreneurship Development Cell (EDC) member, contributing to 30+ startup ideas and innovation initiatives.
- Developing 'Life OS,' a full-stack productivity platform integrating task management, habit tracking, and analytics.